Multi-Agent System: Theories and Applications

Lecture 2: Backgrounds on Matrix Theory

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Outline

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Notations and Symbols

- $\mathbb{R}, \mathbb{C}$: set of real and complex numbers
- $\mathbb{R}_+, \mathbb{C}_+$: set of positive real numbers and complex numbers with positive real parts
- $\mathbb{R}_{0,+}, \mathbb{C}_{0,+}$: set of non-negative real numbers and complex numbers with non-negative real parts
- $\mathbb{R}^n, \mathbb{C}^n$: set of real and complex n-dimension column vectors
- $\mathbb{R}^{n\times m}, \mathbb{C}^{n\times m}$: set of real and complex matrices with dimensions $n \times m$
- A pair $\lambda \in \mathbb{C}, v \in \mathbb{C}^n$ is an eigenvalue and its associated eigenvector of a matrix $A \in \mathbb{R}^{n\times n}$ if $Av = \lambda v$
- A pair $\lambda \in \mathbb{C}, w^T \in \mathbb{C}^{1\times n}$ is a *left eigenvalue* and its associated *left eigenvector* of a matrix $A \in \mathbb{R}^{n\times n}$ if $w^T A = w^T \lambda$
- $\mathrm{Ker}(A)$: the kernel of A, i.e., set of $x \in \mathbb{R}^n$ s.t. $Ax = 0$
- $\mathrm{Im}(A)$: the image of A, i.e., set of $y \in \mathbb{R}^n$ s.t. $\exists\, x \in \mathbb{R}^n : Ax = y$
- $\mathrm{rank}(A)$: the rank of A
- $\mathrm{diag}(\cdot)$: diagonal or block-diagonal matrix
- $|\cdot|$: absolute value
- $\|\cdot\|$: norm, e.g., $\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty$
- $A = [a_{ij}]$: a matrix A with elements a_{ij}

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Location of a matrix's eigenvalues

Consider a matrix $A = [a_{ij}] \in \mathbb{C}^{n\times n}$. Denote $R_i \triangleq \sum_{j\neq i} |a_{ij}|$.

Gershgorin Circle Theorem

Every eigenvalue of A belongs to at least one of the discs $D_i(a_{ii}, R_i)$ centered at a_{ii} with radius R_i.

Note:

- Since eigenvalues of A and A^T are the same, Gershgorin Theorem is also applied for the discs with off-diagonal column sums.

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- Since eigenvalues of A and A^T are the same, Gershgorin Theorem is also applied for the discs with off-diagonal column sums.
- If the union of k discs is disjoint from the union of the other $n - k$ discs then the former and latter union contains exactly k and $n - k$ eigenvalues of A; respectively.

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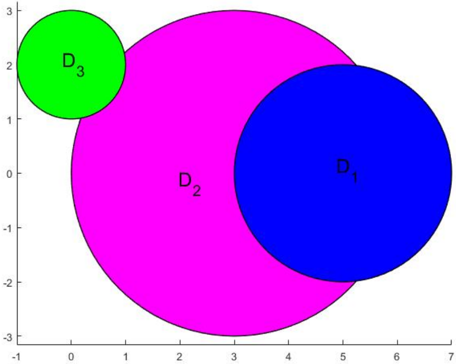
Illustrative example

$$A = \begin{bmatrix} 5 & 2 & 0 \\ 1 & 3 & 2 \\ 1 & 0 & 2i \end{bmatrix}$$

Eigenvalues: $5.9011 + 0.0530i, 2.0136 - 0.2485i, 0.0853 + 2.1955i$



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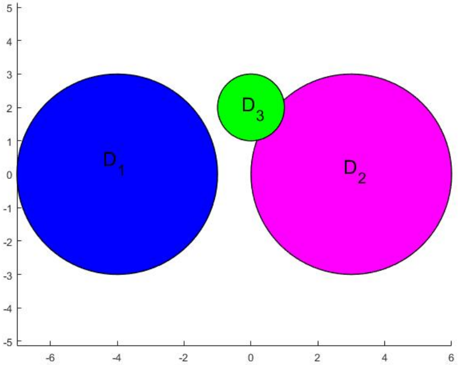
Illustrative example

$$A = \begin{bmatrix} -4 & -3 & 0 \\ 1 & 3 & 2 \\ 1 & 0 & 2i \end{bmatrix}$$

Eigenvalues: $-3.7456 + 0.0996i, 2.3209 - 0.2289i, 0.4247 + 2.1293i$



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Reducibility

A matrix $A \in \mathbb{R}^{n \times n}$ is *reducible* if either (i) $n = 1$ and $A = 0$; or (ii) $n \geq 2$ and there exists a permutation matrix $P \in \mathbb{R}^{n \times n}$ such that $P^T AP$ is a block upper or lower triangular matrix.

A matrix is *irreducible* if it is not reducible. If A is non-negative and $\sum_{k=0}^{n-1} A^k$ is positive, then A is irreducible.

A matrix is *primitive* if it is irreducible and has exactly one eigenvalue of maximum modulus. If A is non-negative and $\exists\ k > 0$ s.t. A^k is positive then A is primitive.

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Example:

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix} : \text{ reducible}$$

$$A_2 = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix} : \text{ irreducible}$$

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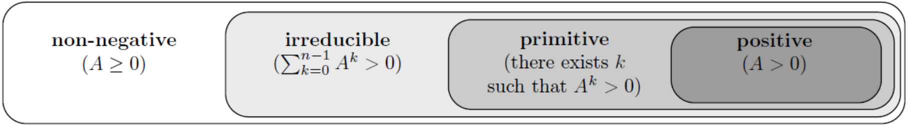
Answer: strong connection to graph theory, shown in the next lecture

Reducibility - Summary



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Periodicity

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Question: How the periodicity is related to the reducibility?

Stochasticity

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Example:

$$A = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 1/3 & 1/3 & 1/3 \\ 0 & 1/3 & 1/3 & 1/3 \end{bmatrix}: \text{ right stochastic, but not left nor doubly stochastic}$$

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Question: How the spectrum of a right stochastic matrix looks like?

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Main Results - Strong Connections to Consensus of MASs

Perron-Frobenius Theorem

Consider a matrix $A \in \mathbb{R}^{n \times n}$, $n \geq 2$. *If A is non-negative, then*

- there exists a real eigenvalue λ, called *the Perron root* or *the Perron-Frobenius eigenvalue*, such that $\lambda \geq |\mu| \geq 0$ for all other eigenvalues μ of A,
- the right and left eigenvectors v and w^T of λ can be selected non-negative.

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If additionally A is irreducible, then

- the eigenvalue λ is strictly positive and simple,
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If additionally A is primitive, then

- the eigenvalue λ is strictly greater than all other eigenvalues, called *dominant eigenvalue*,
- let v and w be normalized, i.e., $w^T v = 1$, then

$$\lim_{k\to+\infty} \frac{A^k}{\lambda^k} = vw^T (\text{ rank-1 Perron projection})$$